

Firm Demographics and Sectoral Reallocation

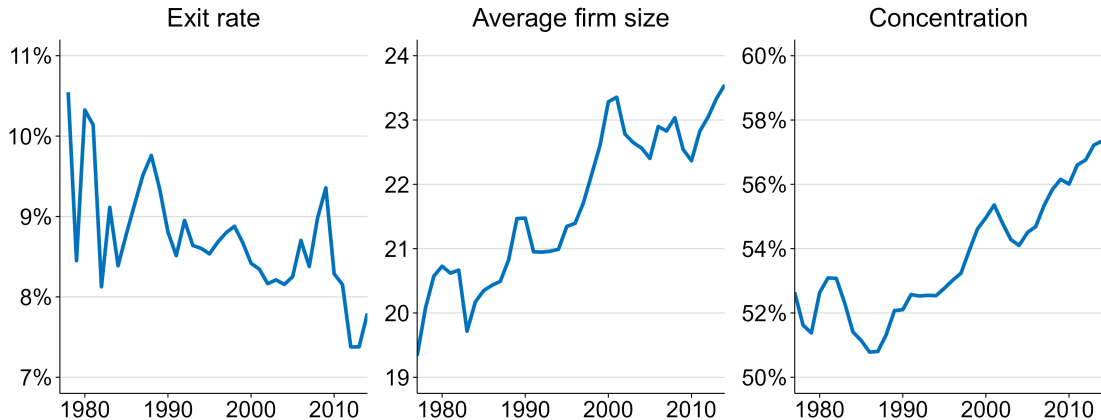
Hopenhayn, Neira, Singhanian (2026, WP)

Miguel Vázquez-Carrero

Macro Reading Group UC3M

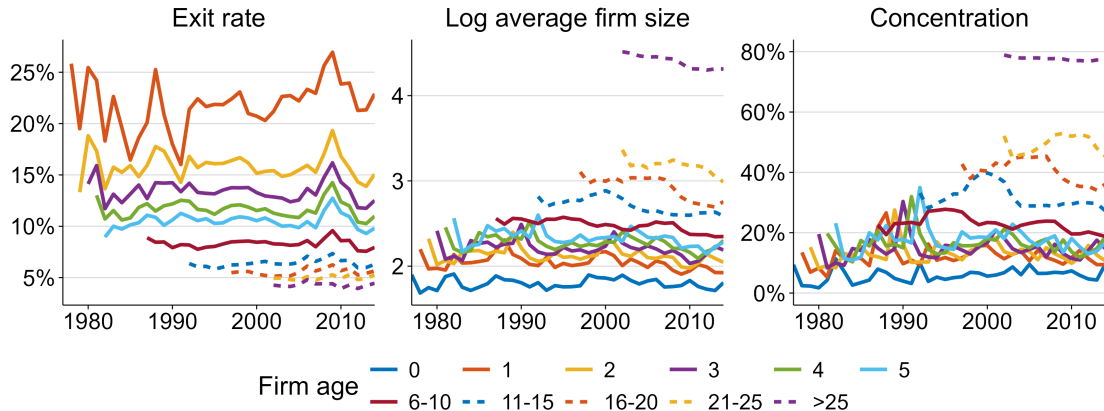
Aggregate Firm Demographics

- Average firm size and employment concentration rise, while the aggregate exit rate falls.



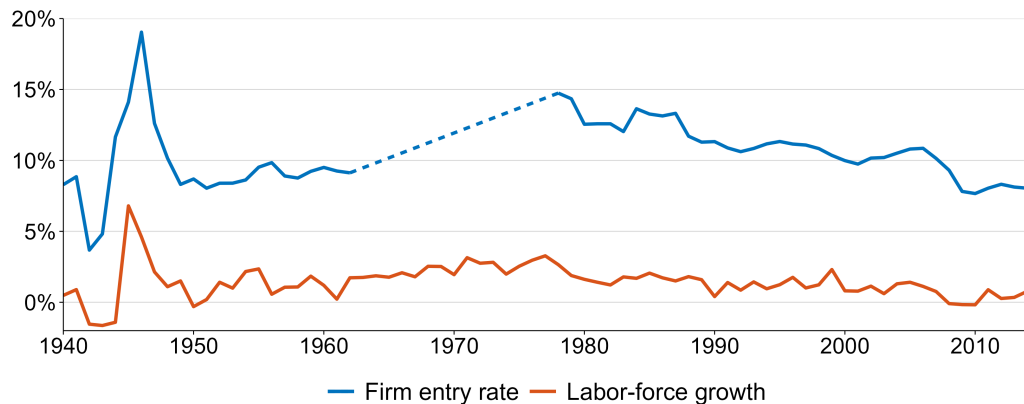
Firm Lifecycle Profiles

- Within firm-age groups, size, concentration, and exit rates are roughly stable over time.



Labor-Force Growth and Firm Entry

- Firm entry and labor-force growth decline together.
- The one-sector model links slower growth to fewer entrants relative to incumbents.



Firm Demographics ($J = 1$)

Timing within Period t

1. **Labor supply is set:** $L_t = (1 + g_t)L_{t-1}$.
2. **Incumbents draw** $z_t \sim F(\cdot \mid z_{t-1})$. Those with $z_t \geq z_t^*$ operate; their total labor demand is L_t^I .
3. **Entry attempts pay** c^e **in labor and draw** $z_t \sim G$. Draws above the same cutoff operate and demand $\ell(z_t, p_t)$; the rest exit.
4. **Entry fills the labor left after incumbents.** The mass m_t is pinned down when setup costs and operating entrants exhaust it:

$$L_t - L_t^I = m_t \left[c^e + \int_{z \geq z_t^*} \ell(z, p_t) dG(z) \right] = m_t \ell_t^e.$$

Firm Demographics: Static Firm Problem

- A firm uses **productive labor** n plus c^f units of **overhead labor**.

$$q(z, n) = z^{1-\gamma} n^\gamma, \quad \ell(z, p_t, w_t) = n(z, p_t, w_t) + c^f$$

- With output price p_t and wage w_t , the firm chooses productive labor to maximize static profits:

$$\pi(z, p_t, w_t) = \max_{n \geq 0} \{ p_t z^{1-\gamma} n^\gamma - w_t n - w_t c^f \}.$$

- Normalize the wage to one: only the relative price p_t/w_t matters.

$$\pi(z, p_t) = \max_{n \geq 0} \{ p_t z^{1-\gamma} n^\gamma - n - c^f \}$$

Firm Demographics: Exit and Entry

- Given the deterministic price path $\mathbf{p}_t \equiv \{p_\tau\}_{\tau \geq t}$, the firm chooses exit or operation:

$$v(z, \mathbf{p}_t) = \max \{0, \pi(z, p_t) + \beta E[v(z', \mathbf{p}_{t+1}) \mid z]\}.$$

- The exit cutoff is the lowest productivity with positive continuation value:

$$z_t^* = \inf \{z : \pi(z, p_t) + \beta E[v(z', \mathbf{p}_{t+1}) \mid z] > 0\}.$$

- An entrant pays c^e , draws $z \sim G$, and follows the same exit rule:

$$v^e(\mathbf{p}_t) = \int v(z, \mathbf{p}_t) dG(z) - c^e.$$

Firm Demographics: Law of Motion

- Let μ_t be the measure of operating firms. For a set A , $\mu_t(A)$ is the mass of firms whose productivity lies in A .
- Entrants draw $z \sim G$; incumbents with inherited $x = z_{t-1}$ draw z_t from F . In the AR(1) specification:

$$\log z_t = (1 - \rho)\mu + \rho \log z_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma_\varepsilon^2).$$

$$\mu_{t+1}(A) = m_{t+1} \left(\int_{z \in A, z \geq z_{t+1}^*} dG(z) \right) + \int \left(\int_{z \in A, z \geq z_{t+1}^*} dF(z \mid x) \right) d\mu_t(x)$$

- Read the incumbent term from the inside out: conditional on inherited x , draw the new state z ; keep only operating firms; then integrate over the existing mass $d\mu_t(x)$.

Firm Demographics: Labor Market Clearing

- The measure μ_t has total operating-firm mass

$$M_t \equiv \int d\mu_t(z).$$

- Inelastic labor supply grows at rate g_t :

$$L_t = (1 + g_t)L_{t-1}.$$

- Labor clearing allocates workers to firms and entry:

$$\begin{aligned} L_t &= \int \ell(z, p_t) d\mu_t(z) + m_t c^e \\ &= \int n(z, p_t) d\mu_t(z) + M_t c^f + m_t c^e. \end{aligned}$$

Firm Demographics: Entry and Labor Clearing

- Labor market clearing can be written as

$$L_t = L_t^I + m_t \ell_t^e.$$

- Incumbent labor reflects surviving firms; entrant labor includes the setup cost and operating labor of successful draws:

$$L_t^I = \int \left(\int_{z \geq z_t^*} \ell(z, p_t) dF(z | x) \right) d\mu_{t-1}(x), \quad \ell_t^e = c^e + \int_{z \geq z_t^*} \ell(z, p_t) dG(z).$$

- Residual labor after incumbents determines current entry:

$$m_t = \frac{L_t - L_t^I}{\ell_t^e} = \frac{(1 + g_t)L_{t-1} - L_t^I}{\ell_t^e}.$$

Constant Prices and Dynamic Entry

- Consider a positive-entry equilibrium with a **constant price**:

$$p_t = p^* \quad \forall t \quad \implies \quad z_t^* = z^*.$$

- Then **age profiles are time-invariant**. Incumbent and entrant labor requirements are

$$L_t^I = \sum_{a \geq 1} m_{t-a} S_a \tilde{\ell}_a, \quad \ell_t^e = S_0 \tilde{\ell}_0 + c^e.$$

- Labor market clearing determines entry:

$$L_t = L_t^I + m_t \ell_t^e, \quad m_t = \frac{L_t - L_t^I}{\ell_t^e} = \frac{(1 + g_t)L_{t-1} - \sum_{a \geq 1} m_{t-a} S_a \tilde{\ell}_a}{S_0 \tilde{\ell}_0 + c^e}.$$

The One-Sector Firm-Aging Mechanism

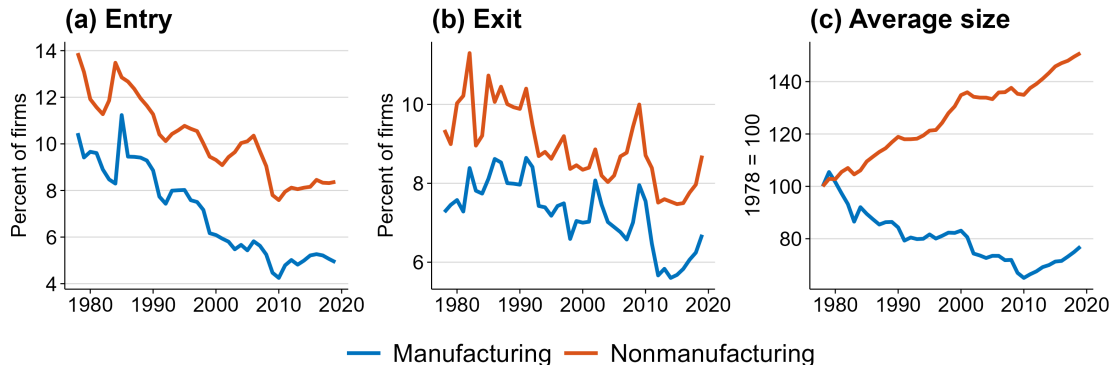
- Slower labor-force growth reduces entry relative to the stock of incumbents.
- Survival shifts the firm-age distribution toward older firms. Their lower exit rates and larger size change aggregate exit and average size.
- Past entry shapes incumbent labor demand L_t^I , which determines labor left for current entry: **own-sector history dependence**.

$$m_t = \frac{L_t - L_t^I}{\ell_t^e}, \quad L_t^I = \sum_{a \geq 1} m_{t-a} S_a \tilde{\ell}_a.$$

Sectoral Reallocation ($J > 1$)

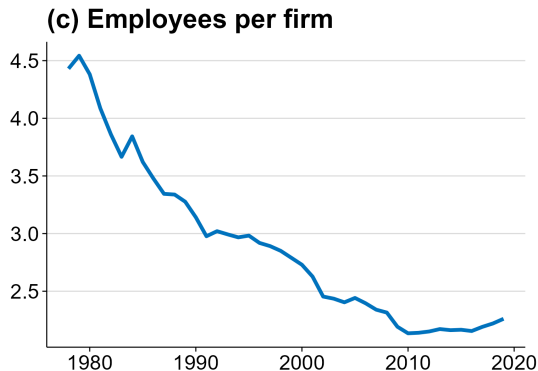
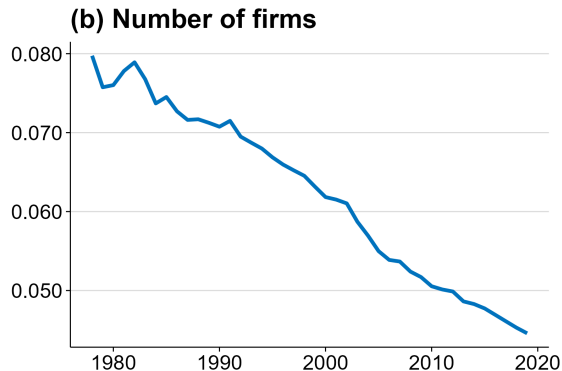
Firm Aging and Sectoral Divergence

- Manufacturing's shares fall: employment 26% $\rightarrow$ 9%; firms 7.4% $\rightarrow$ 4.3%.
- Entry and exit decline in both sectors; average firm size moves in opposite directions.



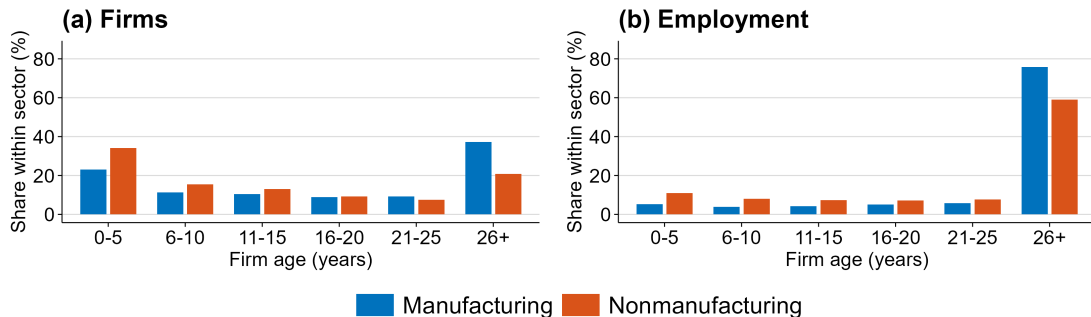
Manufacturing's Relative Decline

- Manufacturing loses firms and employees per firm relative to nonmanufacturing.
- The paper also documents falling relative output and rising relative output per worker.



Firm Age Distribution, 2019

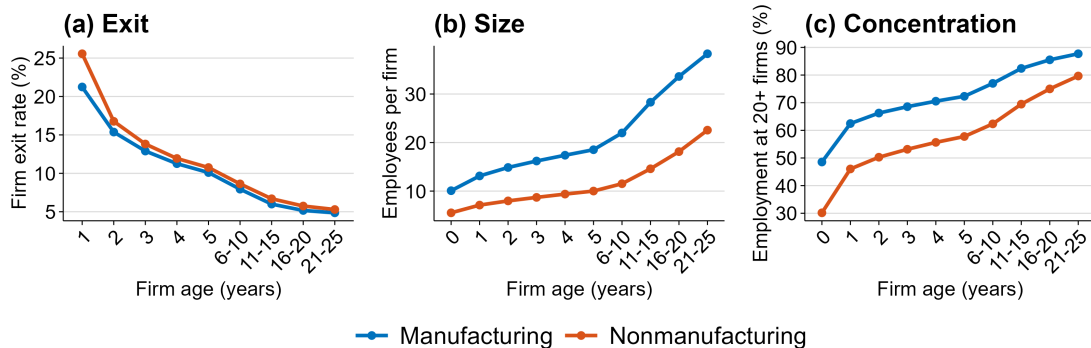
- Manufacturing has fewer young firms: ages 0–5 account for 23% of firms, versus 34% in Nonmanufacturing.



- Firms aged 26+ account for 37% of Manufacturing firms and 76% of its employment, versus 21% and 59% in Nonmanufacturing.

Different Firm Lifecycles Across Sectors

- In both sectors, older firms exit less, grow larger, and concentrate more employment.



- The response to slower population growth depends on these profiles. The paper finds stronger firm-aging feedback and a larger entry decline in Nonmanufacturing.

A Shift–Share View of Firm Aging

- **Share:** $s_{ja,t} = M_{ja,t}/M_{jt}$ is the fraction of sector j 's firms that are age a ; $\bar{q}_{ja,t}$ is their average output per firm.
- **Shift:** slower labor-force growth lowers entry and moves these shares toward older firms. Exit profiles determine how strongly each sector ages.

$$\bar{q}_{jt} \equiv \frac{Q_{jt}}{M_{jt}} = \sum_a s_{ja,t} \bar{q}_{ja,t}, \quad M_{jt} = \frac{Q_{jt}}{\bar{q}_{jt}}.$$

- **Why reallocation follows:** older firms produce more per firm. Different aging paths change $\bar{q}_j$ differently; a given output requirement then calls for different firm counts. Changes in output per worker also move employment shares.

The Firm Problem in Multiple Sectors

- In sector $i = 1, \dots, J$, a firm uses direct labor n and intermediate commodities $\mathbf{x}$ at the **current price vector** $\mathbf{p}_t = (p_{1t}, \dots, p_{Jt})$:

$$q_i(z, n, \mathbf{x}) = z^{1-\gamma_{it}} f_i(n, \mathbf{x})^{\gamma_{it}}, \quad \ell_i = n_i + c_i^f.$$

- With labor as numeraire, total one-period profits are

$$\pi_i(z, \mathbf{p}_t, S_t) = \max_{n, \mathbf{x}} \left\{ p_{it} z^{1-\gamma_{it}} f_i(n, \mathbf{x})^{\gamma_{it}} - n - \mathbf{p}_t \cdot \mathbf{x} - c_i^f \right\}.$$

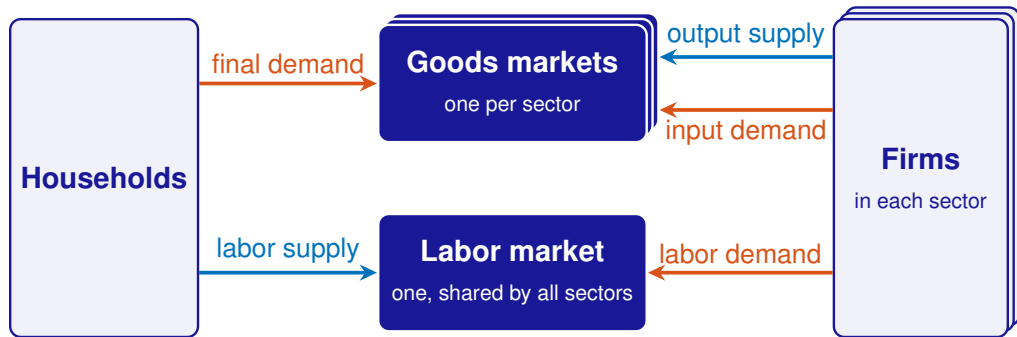
- The Bellman equation keeps exactly the HNS22 form:

$$v_i(z; \mathbf{p}_t, S_t) = \max \left\{ 0, \pi_i(z, \mathbf{p}_t, S_t) + \beta E[v_i(z'; \mathbf{p}_{t+1}, S_{t+1}) \mid z, S_t] \right\}.$$

Demand and Labor Supply Assumptions

- **Labor supply is exogenous:** L_t follows the observed labor-force path.
- **Final-demand schedules are taken as given here:** $d_t(\mathbf{p}_t, E_t)$. The paper derives them from household preferences; its quantitative exercise uses Cobb–Douglas expenditure shares set from the data.
- **Aggregate expenditure is solved in equilibrium:** goods requirements pin down sectoral entry conditional on E_t , and labor clearing pins down E_t .

Goods and Labor Market Participants



Goods and Labor Market Participants



Firm entry adds output supply *and* labor demand at the same time.

Two Market-Clearing Conditions

- **Goods markets, one per sector:** final and intermediate demand require output. Incumbents supply Q_{jt}^I ; entrants fill the gap:

$$Q_{jt}^{\text{req}}(E_t) = Q_{jt}^I + m_{jt}q_{jt}^e, \quad j = 1, \dots, J.$$

- **One common labor market:** incumbents and entry attempts across all sectors use the available labor supply:

$$L_t = \sum_{j=1}^J L_{jt}^I + \sum_{j=1}^J m_{jt}\ell_{jt}^e.$$

- **Joint clearing:** at given prices, goods requirements pin down sectoral entry conditional on expenditure E_t ; the common labor constraint pins down E_t .

Goods-Market Linkages Across Sectors

- Producing one unit in sector i requires $\Omega_{ji,t} \equiv x_{ji}(1, \mathbf{p}_t)/q_i(1, \mathbf{p}_t)$ units of good j as an intermediate input.
- Each goods market must meet final demand and demand from other producers:

$$\mathbf{Q}_t = \Omega_t^\top \mathbf{Q}_t + \mathbf{d}_t(\mathbf{p}_t, E_t).$$

$$\Psi_t \equiv (I - \Omega_t^\top)^{-1} \quad (\text{Leontief inverse}).$$

$$\mathbf{Q}_t^{\text{req}}(E_t) = \Psi_t \mathbf{d}_t(\mathbf{p}_t, E_t).$$

- At a given E_t , demand in sector i raises output requirements in every linked sector through Ψ_t . Those requirements change the sectoral output gaps that entrants fill.

Reading the Leontief Inverse

- $\Omega_{ji,t}$ measures **direct** use of good j by sector i ; $\Psi_t = (I - \Omega_t^\top)^{-1}$ adds the indirect production rounds.

Rows: required output j

Columns: final demand i

		$i = 1$	$i = 2$	$i = 3$
$\Psi_t =$	$j = 1$	$\Psi_{11,t}$	$\Psi_{12,t}$	$\Psi_{13,t}$
	$j = 2$	$\Psi_{21,t}$	$\Psi_{22,t}$	$\Psi_{23,t}$
	$j = 3$	$\Psi_{31,t}$	$\Psi_{32,t}$	$\Psi_{33,t}$

- Cell $\Psi_{ji,t}$ is the total output of good j needed for one extra unit of final demand for good i , including indirect inputs.
- $Q_{jt}^{\text{req}} = \sum_i \Psi_{ji,t} d_{it}$: demand for good i can raise the output gap and entry in linked sector j .

Own-Sector History Dependence

- The input-output network converts final demand into required output in sector j :

$$Q_{jt}^{\text{req}}(E_t) = \sum_{i=1}^J \Psi_{ji,t} d_{it}(\mathbf{p}_t, E_t).$$

- Incumbent output Q_{jt}^I reflects past entry and survival in sector j ; each entry attempt adds q_{jt}^e units of output.
- Sector- j entry fills the remaining **output gap**:

$$m_{jt} q_{jt}^e = Q_{jt}^{\text{req}}(E_t) - Q_{jt}^I.$$

Cross-Sector History Through the Labor Market

- The common labor constraint pins aggregate expenditure E_t after accounting for incumbent output and labor:

$$\sum_{j=1}^J \frac{\ell_{jt}^e}{q_{jt}^e} \left(\sum_{i=1}^J \Psi_{ji,t} d_{it}(\mathbf{p}_t, E_t) \right) = L_t - \left[\sum_{j=1}^J L_{jt}^I - \sum_{j=1}^J \frac{\ell_{jt}^e}{q_{jt}^e} Q_{jt}^I \right].$$

- Past entry and survival in every sector determine the incumbent terms Q_{jt}^I and L_{jt}^I .
- A shock to sector k changes E_t and current entry in every other sector $j \neq k$.

Lifecycle Differences Reallocate Firms and Workers

- Population slowdown alone: entry falls, firms age, relative output stays fixed.

$$\frac{M_{it}}{M_{jt}} = \underbrace{\frac{Q_{it}}{Q_{jt}}}_{\text{relative output}} \times \underbrace{\frac{Q_{jt}/M_{jt}}{Q_{it}/M_{it}}}_{\text{output per firm } (j/i)},$$

$$\frac{N_{it}}{N_{jt}} = \underbrace{\frac{Q_{it}}{Q_{jt}}}_{\text{relative output}} \times \underbrace{\frac{Q_{jt}/N_{jt}}{Q_{it}/N_{it}}}_{\text{output per worker } (j/i)}.$$

- The output ratio stays fixed here. The second factor moves as each sector's age mix changes; older firms spread fixed overhead labor c_i^f over more output.

$\downarrow g_t \Rightarrow$ relative firms and employment shift toward Manufacturing.

From One Sector to Two: Quantitative Application

- **One sector** ($J = 1$): entry depends on labor left after incumbents, capturing own-sector history.

$$m_t = \frac{L_t - L_t^I}{\ell_t^e}.$$

- **Two sectors** ($J = 2$): Manufacturing (M) and Nonmanufacturing (NM) retain this entry condition, linked by intermediate inputs, cross-sector history, and a common labor market.
- **Quantitative drivers:** population/labor-force growth g_t ; sectoral technology and final-demand preferences with expenditure share b_t ; and international trade.

Trade and Domestic Output Requirements

- Trade wedges U.S. usage from required domestic output. Let U_{jt} be total usage, ϕ_{jt} its import share, and χ_{jt} the export share of gross output.

$$(1 - \chi_{jt})Q_{jt} = (1 - \phi_{jt}) \left[D_{jt} + \sum_i \int X_{ji,t}(\mathbf{p}_t, z) d\mu_{it}(z) \right].$$

$$U_{jt} \equiv D_{jt} + \sum_i \int X_{ji,t}(\mathbf{p}_t, z) d\mu_{it}(z), \quad Q_{jt} = \frac{1 - \phi_{jt}}{1 - \chi_{jt}} U_{jt}.$$

- When ϕ_{Mt} rises, imports supply more U.S. manufacturing usage, so less domestic manufacturing output is required.
- The smaller output gap reduces Manufacturing entry and ages its firms; age-specific size and exit profiles remain fixed:

$$\downarrow Q_{Mt}^{\text{req}} \Rightarrow \downarrow (Q_{Mt}^{\text{req}} - Q_{Mt}^I) \Rightarrow \downarrow m_{Mt}.$$

Sectoral Demographic Spillovers

- As firms age, sector-specific lifecycle profiles change incumbent labor demand.
- The common labor market transmits those changes to entry in both sectors:

$$\underbrace{\sum_{j=1}^J m_{jt} \ell_{jt}^e}_{\text{labor for entrants}} = L_t - \underbrace{\sum_{j=1}^J L_{jt}^I}_{\text{labor at incumbents}} .$$

- The two lifecycle profiles push Manufacturing entry in opposite directions:

Manufacturing age profile

Without it, the model predicts too small a decline in manufacturing entry.

Nonmanufacturing age profile

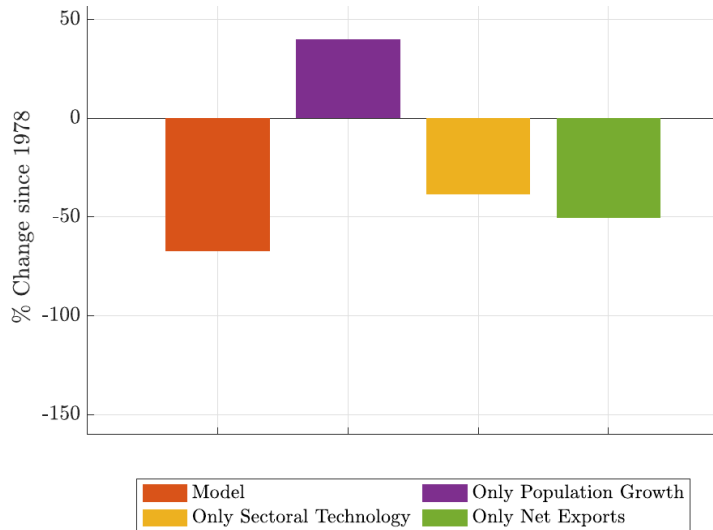
Without it, nonmanufacturing absorbs more labor, and manufacturing entry falls too much.

Application: Manufacturing vs. Nonmanufacturing

- **Population growth:** slower growth lowers entry, especially in Nonmanufacturing. Stronger aging there raises average size and, on its own, shifts firms and employment *toward* Manufacturing.
- **Technology and preferences:** Manufacturing becomes less labor-intensive and relative demand moves away. Employment falls *within age bins*; technology explains most of its employment decline.
- **Trade:** a rising net import share reduces domestic Manufacturing output and entry. Firms age, and the Manufacturing firm share falls.

Technology and trade outweigh the demographic pull.

Driving Forces of Manufacturing Reallocation



- Trade lowers Manufacturing's relative firm count.
- Slower population growth offsets part of this decline by raising Nonmanufacturing's average size.

Journal of Monetary Economics

Date: August 2019

Show more ✓

Research article ● Full text access [Get rights and content ↗](#)

Taxation and the life cycle of firms ☆

Andrés Erosa^a  , Beatriz González^{a b} [Show more](#) ✓

A Happy Idea for my JMP?

- **Erosa–González:** use Hopenhayn to study how different ways of taxing capital income affect firms' lifecycles and entry.
- **Uzawa:** the classic two-sector model of investment and consumption goods.
- **JMP question:** how does taxation change firm growth and lifecycle dynamics with input–output linkages and different profiles across sectors?
- **HNS:** a multisector firm-dynamics framework to start answering this question.